MATH110BH Homework 4

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1 Problem 5

Lemma 1.1. Let R be a domain. If every module over R is free, R is a field.

Proof. Suppose not. Then, R has a non-zero non-unit ideal I. Consider the R module R/I. But free modules over domains have to be torsion-free and therefore R/I isn't free. $\square$